Estimating Nonlinear Insulin-Glucose Dynamics in Critically-ill Patients

Markus Mulvihill

KTH Royal Institute of Technology

August 7th, 2026

2026-07-23

Thesis Presentation

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Stress Hyperglycaemia in ICU Patients

1. Patients suffering from critical illnesses are subjected to metabolic changes from acute stress

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1. Inflammatory response secretes stress hormones
2. Acute myocardial infarction, sepsis, stroke

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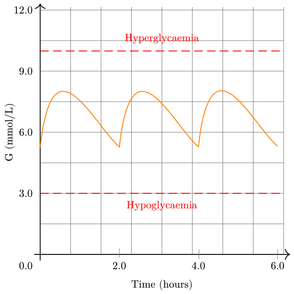
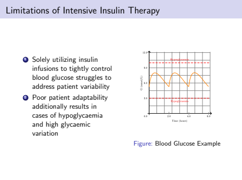


Figure: Blood Glucose Example



1. Infrequent measurements, not considering nutrition, and not complex to monitor dynamics changes in patient health
2. A patient with exogenous insulin that exponentially decreases

Role of Model-based Glycaemic Control

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1. Better method than IIT to be specific

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Intensive Control Insulin-Nutrition-Glucose (ICING) model

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References

Berghe, Greet Van den et al. (Nov. 2001). “Intensive Insulin Therapy in Critically Ill Patients”. In: DOI: 10.1056/NEJMoa011300.

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References

Berghe, Greet Van den et al. (Nov. 2001). “Intensive Insulin Therapy in Critically Ill Patients”. In: DOI: 10.1056/NEJMoa011300.